Applied Energy 253 (2019) 113538

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

Numerical analyses of hybrid jet impingement/microchannel cooling device for thermal management of high concentrator triple-junction solar cell

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HIGHLIGHTS

- A three-dimension model for high concentrator photovoltaic (HCPV) was developed.
- The jet impingement/microchannel hybrid cooling scheme was studied for HCPV.
- The hybrid schemes have a higher thermal performance and overall exergy efficiency.
- Thermal stresses sufficiently decreased with increasing the inlet mass flow rate.

ARTICLE INFO

Keywords:
HCPV/T
Multijunction solar cell
Jet impingement/microchannel
Exergy
Structure analysis

ABSTRACT

An efficient cooling arrangement is mandatory to achieve a higher net output power from the high concentrator photovoltaic structures in addition to extending their lifetime. In the current study, five new heat sink designs for a jet impingement/microchannel hybrid cooling scheme were investigated and compared with a conventional jet impingement cooling scheme. These designs consisted of an arrangement of rectangular fins at the streamwise length of the heat sink. This resulted in a stepwise decrease in the corresponding channel width and hydraulic diameter. A comprehensive three-dimensional thermal and structure model was developed to investigate the capability of the proposed designs in terms of reduction of the cell temperature besides enhancement of the temperature uniformity. Based on the results, the hybrid cooling scheme exhibited promising cooling ability compared to the conventional jet impingement scheme. The results of the present study show that the hybrid cooling scheme is effective cooling system and it achieved the utmost possible reduction of solar cell temperature, under high solar concentration ratio of 1000 suns where the solar cell temperature reduces to 55 °C. When the inlet mass flow rate was increased to 50 g/min under the same conditions, a corresponding reduction in the cell temperature from 67.3 to 55 °C was observed for Case 4 of the hybrid scheme designs. In addition, there was a decrease from 82.3 to 63.2° C for Case 1 of the conventional jet impingement heat sink (HS). Under the hybrid cooling scheme, the electrical efficiency of the cell improved to 39.7% when the inlet mass flow rate was equal to 50 g/min for Case 4. Exergy analysis revealed that the hybrid scheme achieved an overall exergy efficiency of 53.5% at inlet mass flow rate of 25 g/min.

1. Introduction

Solar energy is an attractive alternative as a green, eco-friendly, and unlimited energy resource [1]. The conversion of the solar energy to thermal and electrical energy represents the most common application of solar energy usage [1]. Photovoltaic (PV) solar cells are the most ubiquitous and promising solar energy technology.

Nonetheless, solar energy exploitation for electrical power generation is still limited as the technology is not yet economically competitive in comparison with the conventional energy sources [2]. High

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https://doi.org/10.1016/j.apenergy.2019.113538
Received 5 December 2018; Received in revised form 16 June 2019; Accepted 11 July 2019
Available online 20 July 2019
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Nomenclature

A	cell area (m^2)
CR	concentrations ratio (Suns)
DNI	direct solar radiation (W/m^2)]
E	exergy (W)
h	convective heat transfer ($W/m^2 \cdot K$)
H	channel height (m)
k	thermal conductivity (W/m·K)
L	length (m)
ṁ	mass flowrate (kg/s)
P	pressure (Pa) or electrical power, friction power and net power (W)
q	heat flux (W/m^2)
Q	thermal power (W)
R	thermal resistance (K/W)
T	temperatures (°C)
V	solar cell volume (m^3)
v	velocity (m/s)
W	width (m)

Greek symbols

μ	water viscosity (Pa·s)
α	the thermal expansion coefficient or absorptivity
$\beta_{thermal}$	temperature thermal coefficient (1/K)
Δ	difference
δ	thickness (m)
ε	thermal strain (mm/mm)
η	efficiency (%)
ξ	exergy efficiency (%)
ρ	water density (kg/m^3)
σ	thermal stress (MPa)
Ψs	exergy solar conversion coefficient

Subscript

a	ambient
cell	solar cell
ch	channel
conv	convection
cu	copper
el	electrical or elastic
f	fluid
Ge	germanium
HS	heat sink
in	inlet
j	jet
out	outlet
rad	radiation
Ref	reference
th	thermal
W	wall
Wind	Wind
fric.	friction

Abbreviations

PV	photovoltaic
MJ	multijunction
JIMC	jet impingement/microchannel
JI	jet impingements
HS	heat sink
HCPV/T	high concentrator photovoltaic/thermal system
HCPV	high concentrator photovoltaic
CPV	concentrator photovoltaic
3J	triple junction

concentrator photovoltaic (HCPV) systems present a potential solution for reducing the cost of the PV systems [3]. The concentrating units can focus an enormous amount of solar energy on small solar cells. Thus, the active area of the expensive semiconductor can be replaced by low-cost optics arrangements. The optical concentrator system could be parabolic troughs, dishes, Fresnel lens, or some other optical system [4]. Also, the HCPV systems usually consist of multi-junction (MJ) solar cells which have higher efficiencies along with a small temperature performance dependance. The MJ solar cells can work efficiently under high values of solar concentration ratio (CR) up to 3000 [5]. Lately, it has been reported that the electrical efficiency of these cells could be up to 46%, and it is anticipated that this value will improve to 50% by the next decade [3].

Despite the considerable progress in these solar cells, a large portion of the incident sunlight is still converted into heat, which leads to a high cell temperature (T_{cell}) values that decrease the efficiency of the cell and could cause damages [6]. Additionally, the existence of high T_{cell} and hot spots lead to physical degradation in the cell structure due to thermal stresses and thermal fatigue [6]. Indeed, sustaining the cell under standard operating temperature range is the primary challenge of HCPV systems to produce electricity at the ultimate performance in all situations. An extensive literature on the impact of T_{cell} on solar cell performance is available in [7]. Landis et al.[8] studied the temperature dependence of 3J solar cells under low solar irradiance and low-temperature conditions in the temperature range of −160 to 20 °C. Lv et al. [9] developed a comprehensive model to estimate the temperature-dependent of 3J solar cells under CR = 576 Suns. Helmers et al. [10] investigated 3J sub-cells individually and studied the temperature dependence as well as the influence of temperature on current-matching. This investigation was conducted for a wide range of operating conditions including cell temperatures from 5 to 170 °C and CR = 1 to 3000 Suns. Nishioka et al. [11] presented a correlation for the 3J solar cell conversion efficiency and cell temperature coefficient. They indicated that the 3J solar cell conversion efficiency reduced by 0.248% per rising in the T_{cell} at CR = 1 Sun and by 0.098%/K at CR = 200 Suns. For the latest 3J version of AZURESPACE data sheet [12], the open circuit voltage (V_{oc}) of a cell is reduced by 0.135%/K under CR = 500 Suns. Therefore, thermal management of HCPV is essential in maintaining efficient operation and avoid irreversible degradation resulted from long-term high operating temperatures [6].

Once, an ineffective cooling is applied for HCPV; the cell temperature gradient occurs to some extent. The temperature gradient usually associates with thermal stress in the cell body. Thermal stress is the primary challenge for the HCPV structure as it is a negative aspect [13]. The literature indicates that some recent efforts have been directed at permanent structural damage such as deformation, delamination, and microcracks formed by high cell temperatures [14]. The structural damage could happen within the cell assembly when the cell is exposed to high temperature. Some reports considered thermal stress produced in polycrystalline PV cells and organic PV cells [15]. However, very insufficient work is available on the mechanical analyses for MJ cells.

The active cooling arrangements are an efficient, practical, and economically reasonable cooling scheme for HCPV systems [16]. Furthermore, several studies have suggested the removed heat energy by the coolant could be collected and reused. In this case, the HCPV is called HCPV/Thermal system or HCPV/T. Aldossary et al. [17] considered a cooling channel for the HCPV under CR = 500 Suns and proposed a probable coupling of HCPV with an absorption heat pump.

Dong et al. [18] mathematically studied the cooling performance of a T-shaped cooling device for 100 MJ cells. They stated that T-shaped heat sink (HS) could effectively reduce the cell temperature and provide a reasonable temperature uniformity of 26 °C. The principal disadvantage of large channel HS is the large area required for installation [19]. The tracking systems through a confined space for such HCPV/T arrangements represent a crucial challenge [20]. Micheli, et al.[21,22] presented experimental and numerical results and concluded that straight silicon micro-fins HS was able to cool HCPV under CR = 500 Suns to the recommended operating conditions. Di Capua H et al. [19] numerically simulated a microchannel HS with complex geometric for HCPV under *CR* = 1000 Suns. The results showed that microchannel HS devices could maintain the T_{cell} below 28 °C. Radwan et al. [23] compared the solar cell temperature using different cooling techniques available in the literature at different low CR values. Based on this comparison, it was noticed that using microchannels accomplished the maximum reduction in the solar cell temperature. Also, it was concluded that using traditional wide microchannel heat sinks, the cell temperature gradient increases. And increasing the coolant flow rate is the only way to decrease the cell temperature gradient in this design [24]. However, the increase in the flow rate is associated with an increase in pumping power consumption. Therefore, Radwan and Ahmed [25] compared the use of five different microchannel heat sinks to enhance the cell temperature uniformity and decrease the cell temperature gradient with a decrease in the pumping power. They concluded that using non-conventional heat sink like double layers microchannel heat sink achieves the best temperature uniformity with a considerable reduction in the pumping power consumption. On the other hand, the constructal theory introduced by Bejan [26] was adopted by several studies to design multistage bifurcations microchannel or stepwise varying width microchannel (SVWM) heat sinks to maximize the heat transfer rates. Shen et al. [27] numerically optimized the position of internal vertical bifurcations along the streamwise direction of the microchannel heat sink. A 27.36% reduction of thermal resistance was obtained at the highest inlet velocity of 1.88 m/s without any pressure drop penalty as compared with the straight rectangular microchannel. Riera e al. [28] designed and studied SVWM device to improve the temperature uniformity of cooled microelectronics components. Low thermal resistance coefficient of 2.35×10^2 m^2 K/W was obtained at an inlet mass flow rate of 1.6 ml/s.

Meanwhile, many studies have directed much attention to confined and unconfined jet impingement (JI) cooling arrangements, in view of the profile of jets and various coolants under either laminar and turbulent regimes under high heat fluxes [29]. Royane et al. [30] discussed the potential of the JI cooling for HCPV systems. The JI cooling scheme could dissipate a large quantity of heat as very thin heat transfer boundary layers usually develop in the impingement zone directly under the jet. Wang et al. [29] Experimentally studied and compared novel finned copper foam HSs with conventional finned HSs exposed to JI cooling. The influences of HS height, the pores compactness of the copper foam and the gas flow Reynolds number were investigated in the term of thermal and fluid flow performance. They concluded that including copper foam had a positive enhancement of the thermal performance of finned HS exposed to JI. However, a higher-pressure drop was reported higher than the conventional finned heat sinks. Both microchannel and JI cooling systems provide high flux extraction but still have serious deficiencies. For microchannel HS, a high rise in temperature along the flow stream, high power of pumping, and flow maldistribution represent some of the challenges of multichannel cooling devices [25]. Otherwise, in case of JI, many jets are required to guarantee a relatively upright level of uniformity of temperature across the HS base. As a hotspot or the highest local cell temperature generally occurs in the central area of the MJ solar cell even when advanced Jet Impingement (JI) or multichannel cooling devices were applied [18]. The authors' prior work [31] confirmed that the highest local cell temperature in the cell center reached up to 1360 °C on the uncooled solar cell under *CR* = 1000 suns. Therefore, attaining a uniform temperature distribution on the cell is an essential objective of this study by adopting an unconventional active cooling arrangement for the HCPV's cells. To date, several works have been published that consider the microchannel and a jet impingement/microchannel (JIMC) systems in which the benefits of both schemes are united to overcome the significant weaknesses of the two systems.

As a portion of the received solar irradiance is converted to electricity in the MJ solar cells by PV effect, and the remaining portion is absorbed as heat energy which can be utilized for many downstream applications. This absorbed heat energy cannot be harvested as a valuable work till an appropriate temperature difference happens between the high-temperature heat source and the HS with low temperature [32]. The exergy analysis generally is performed to estimate the available energy to provide a guiding principle for more practical system arrangement with improved energy recovery efficiency. Hence, a supplementary characteristic qualitative overall performance evaluation of the HCPV/T systems could be obtained with exergy analysis. Accordingly, exergy analysis was performed in this study to assess the obtainability of thermal energy collected by the coolant and the capability to exploit it for other subsystems [33]. Although the microchannel and jet impugnment heat sinks have been applied separately for the cooling of HCPV systems, a combination between the microchannel and jet impingement heat sink in one design could allow the system to get the advantage of both schemes and not comprehensively investigated in the literature.

Therefore, this study aims to provide an efficient cooling method for the HCPV systems. This cooling method is able to achieve lower solar cell temperature, high- cell temperature uniformity, lower power consumption, and less thermal stress. Accordingly, the originality of the present study is based on two key concepts. First, a new cooling method for HCPV under high solar concentration ratio (CR) of 1000 suns is proposed. In this cooling method, a comparison between two different designs of conventional jet impingement heat sinks and five new distinct designs of hybrid jet impingement microchannel heat sinks is conducted. The comparison is conducted at the same coolant flow rate, channel height, solar CR, and meteorological conditions. Second, thermal, exergetic, and structural analyses for the multijunction solar cell under the proposed cooling methods are performed. To accomplish this target, a comprehensive 3D conjugate heat transfer model for the fluid flow in the new designed heat sinks coupled with the thermal model for the multijunction solar cell is developed. The model is numerically solved and validated with the data in the literature. The suggested hybrid heat sinks will significantly assist in improving the performance of HCPV systems, avoid the hot spot formation in solar cells, and limits the thermal stresses due to the low-temperature gradient across the multijunction solar cell.

2. Physical model

Comparing the thermal, structural, and exergy performance of the proposed HCPV system under different heat sink designs was performed at the same heat sink total area that covers the copper II substrate, the same coolant flow rate, the same solar *CR*, and the same meteorological conditions such as the ambient temperature, wind speed, and solar radiation. The physical assembly of the investigated HCPV/T adopted in the present work is revealed in Fig. 1. To achieve the maximum benefit of the solar irradiance as a renewable source of energy, an HCPV structure was equipped with dual axis tracking system. Fresnel lenses were used to concentrate the solar irradiance onto small area of the 3J solar cell. In this type of solar concentrator, a *CR* up to 1000 suns (one sun = 1000 W/m^2) can be achieved at a low cost [34]. The utilized solar cell has an area of 10×10 mm^2. A typical AZURESPACE solar cell was modeled in the current work. The recommended operating temperature range is between 20 and 80 °C, and the extreme allowable T_{cell} must be less than 110 °C [12]. The HCPV structure consisted of a cell,

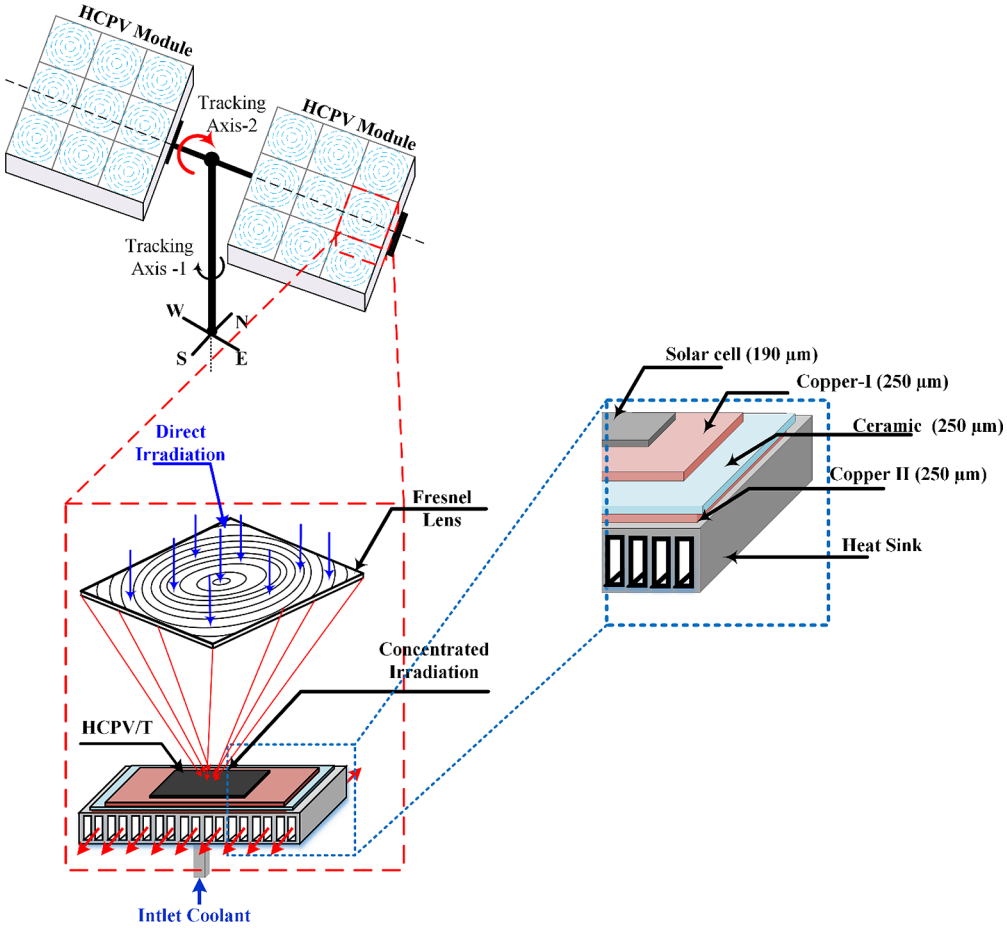


Fig. 1. Assembly of the proposed HCPV/T arrangement cooled with hybrid JIMC heat sink.

Table 1
Detailed solar cell dimensions (mm).

Layer	Thickness (δ_{Hs})	Length (L)	Width (W)
Cell	0.19	10	10
Copper-I	0.25	24	19.5
Al_2O_3 ceramic	0.32	25.5	21
Copper-II	0.25	25	20.5
HS target wall	0.5	25.5	21

two copper sub-layers, and a ceramic layer [12]. The cell and board structure dimensions layers are tabularized in Table 1.

In the present 3J solar cells, the copper and ceramic board, for thermal and electrical power handling, have a length around 2.5 times of the cell layer. The red dashed square in Figs. 2 and 3 indicates the cell location concerning the heat sink. Therefore, heat dissipation from the cell layer to the substrate layers occurs. In case of no cooling, the highest temperature occurs at the middle of the cell layer due to the heat dissipation from the substrate layers as will be presented later.

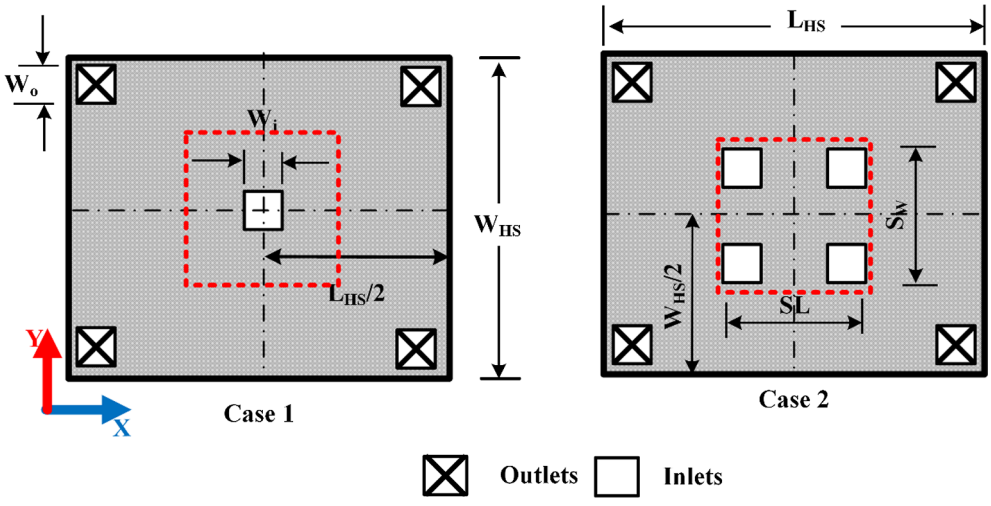


Fig. 2. Inlets and outlets location for JI scheme (Case 1 and Case 2).

Table 2
Dimensions of the investigated JI designs.

Dimension	Value (mm)	Dimension	Value (mm)
W_{HS}	25.5	W_j, W_o	1
L_{HS}	21	S_L	8.67
H_{HS}	4	S_w	7.16

Recently, it was noticed that using the conventional wide microchannel heat sink, the coolant passes from one side of the solar cell and exist for the other side. Therefore, the flow path has the same length as the copper II substrate layer. In this case, the coolant temperature increases with the flow direction and the maximum cell temperature will be shifted toward the side where the coolant outlet exists. In this case, the extended flow path could increase the temperature gradient on the solar cell. To avoid such problem, two conventional designs of jet impingement heat sinks were proposed as shown in Fig. 2. In these cases, the flow path is shorthand. Also, the minimum coolant temperature at the inlet is just directed to the hottest place just underneath the middle of the germanium layer. Therefore, a higher reduction in the maximum temperature could be achieved. Not only this but also for the same flow rate, the coolant velocity inside the channel will be lower. Therefore, the friction power reduction is expected. The use of four inlet jets is also compared with the single jet. The best of them is selected for the next new five designs. In more details, two different designs of a conventional JI were investigated and compared with five different JIMC designs. The studied JI designs are shown in Fig. 2. The first design of the JI included one inlet (Case 1). The second design of the JI consists of four inlets (Case 2). All the HS designs dimensions are listed in Table 2. The (H_{HS}/W_j) ratio was selected to be 4 based on [30].

The proposed JIMC cooling scheme cases (Case 3 to Case 7) are presented in Fig. 3. All the proposed designs included one inlet as previously discussed for Case 1. The one jet inlet was selected based on the results obtained from previous works of the authors [31]. The fins (0.5 mm wide) served as vertical walls in the microchannels. In the geometric design, L, W, and H denote the length, width, and height, respectively. All the geometric parameters are listed in Table 3. The cooling liquid passed in the HS over the single jet and impinges in the central zone of the HS top wall. At that moment, the fluid was separated into two opposite streams and flowed through variable-width microchannel zones.

In JIMC hybrid scheme cases, the values of the channel width were designed to be ranged from 3.5 mm to 0.5 mm, and the impinging zone was in the central area of the HS. The channel widths varied and were arranged at different points along the cooling liquid stream as previously mentioned in the introduction section. For more details, different channels widths varied at separate section along the cooling liquid stream, ranging from 3.5 mm to 0.5 m. The high-velocity coolant

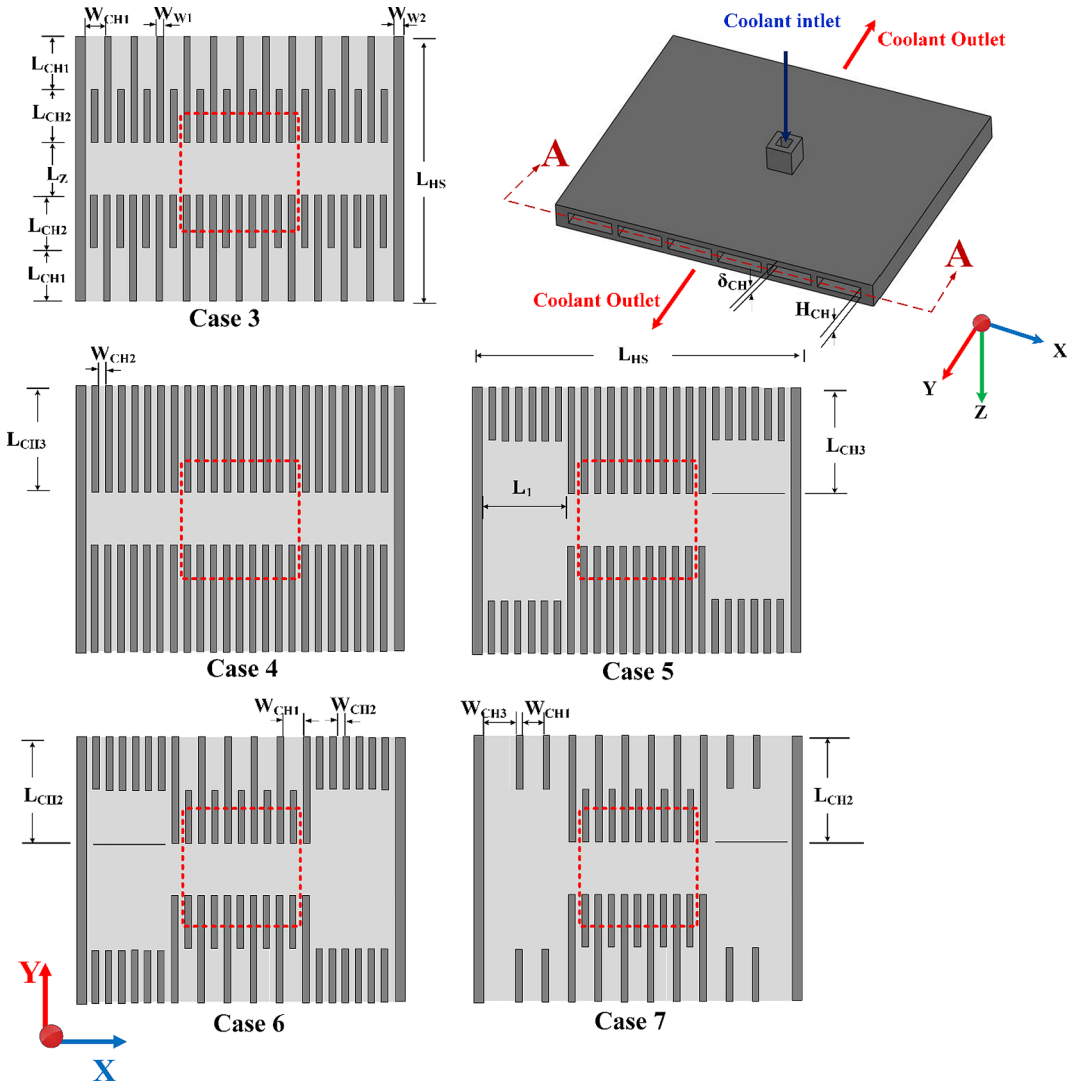


Fig. 3. Detailed dimensions of the investigated JIMC hybrid cooling heat sinks scheme cases.

Table 3
Detailed dimensions of the investigated JIMC designs.

Dimension	Value (mm)	Dimension	Value (mm)
W_{HS}	25.5	W_{W1}	0.5
L_{HS}	21	W_{W2}	0.75
H_{CH}	2	L_1	6.5
δ_{Hs}	0.5	L_Z	4.1
W_{CH1}	0.5	L_{CH1}	4.35
W_{CH2}	1.5	L_{CH2}	4.1
W_{CH3}	3.5	L_{CH3}	8.45

formed a jet that impacts the center of the HCPV backside. The cooling fluid stream than was literally forced to be driven from both sides of the impingement zone through microchannels section. The channel design for each case was adjusted according to HCPV heat flux distribution (the red rectangle in Fig. 3 indicates the hotspot location), the pumping requirements and fabrication limitations. As shown in Fig. 3, the flow free space in the heat sink in Case 1 and Case 2 was replaced with multi-channels with different designs in the new design cases of the JIMC hybrid heat sink. The idea of using multichannel is favorable compared to the wide channel, since the ribs, solid regions between two adjacent flow channels, act as a thermal bridge to transfer the heat quickly from the copper II substrate to the coolant from each side. Therefore, better thermal performance is expected. But dividing the flow passage into multi-channels increases the pressure drop and the longer channel length, the higher pressure drops. Furthermore, the smaller channel width is also increasing the friction power and enhance the heat transfer rate. Therefore, a hybrid combination of the heat sink design is proposed in Fig. 3.

The main difference between Case 3 and Case 4 is that stepwise varying width concept was considered in case 3 while not in case 4. In the stepwise varying width, the channel width was decreased under the cell location where high heat transfer is essential, and the channel width was increased at the locations where no high heat flux exists. This could help to decrease the pumping power consumption in Case 3 compared to Case 4 along with less manufacturing time and material. On the other hand, in Case 5, Case 6 and Case 7, the channels are proposed to be manufactured with more considerable channel length under the cell location only while the channels are shorter in the locations away from the germanium layer. This help to further reduce the manufacturing time and the materials used by the 3D printer [35]. Therefore, the manufacturing cost of the designs in Case 7 is lower than Case 6 and lower than Case 5. Among all the proposed designs, Case 7 has the most economical in the manufacturing process, and lees consumed powder in the 3D printing.

3. Numerical analysis

The thermal modeling was conducted using an academic licensed version of ANSYS Fluent. Also, the structure Finite Element Analysis (FEA) was performed using the mechanical simulation tool in ANSYS. The steady state temperature distribution was exported from the thermal modeling for the mechanical simulation. ANSYS Fluent simulation tool was selected as it covers a wide range of the martial library, boundary conditions, solution methods, and discretization schemes. In ANSYS-Fluent tool, the well-known fluid flow equations including continuity, momentum, and energy were the standard equations and could be easily coupled with the usual energy equation for the solid regions as in the current problem needs. Also, this tool contains the most common types of boundary conditions. These boundary conditions can be either constants or a user-defined function.

In the present work, the fluid continuity, momentum, and energy equation for both fluid and solid were the standard equations and chosen to be solved. However, the solution of the energy equation for the cell contained a heat generation term due to the solar light absorption. This term was added to the standard energy equation for cell domain. Also, the heat generation term was not constant as standard exists in Fluent; it is a function of the solar cell temperature which is unknown. Hence, an iterative technique was applied as detailed explained in in the author earlier work [31] Further, the water thermophysical properties were not considered a constant as in the standard Fluent interface. In this case, water density, viscosity, thermal conductivity, and specific heat were considered as a fourth order polynomial functions of the temperature. Further, to avoid reaching the boiling point of the coolant at any cell in the computational domain, the maximum coolant temperature was monitored during the solution for all simulated cases and checked to be less than the water boiling point at the atmospheric pressure.

3.1. HCPV system thermal modeling

It is commonly accepted to simulate the cell as one layer of germanium (Ge) [36]. Chou et al. [37] confirmed that the heat energy is mainly generated in the Ge layer because long-wavelength photons ($1879 > \lambda > 954$ nm) is generally absorbed within the Ge substrate. As shown in Fig. 4, the assembly of the investigated HCPV/T system mainly consists of several layers such as Ge, upper copper I, Al_2O_3-Ceramic, lower copper II, and the proposed HS designs. This section describes the 3J solar cell layers and the conjugate heat transfer model for the HCPV. The assumptions implemented in this study are the same as those mentioned in the authors' prior work [31].

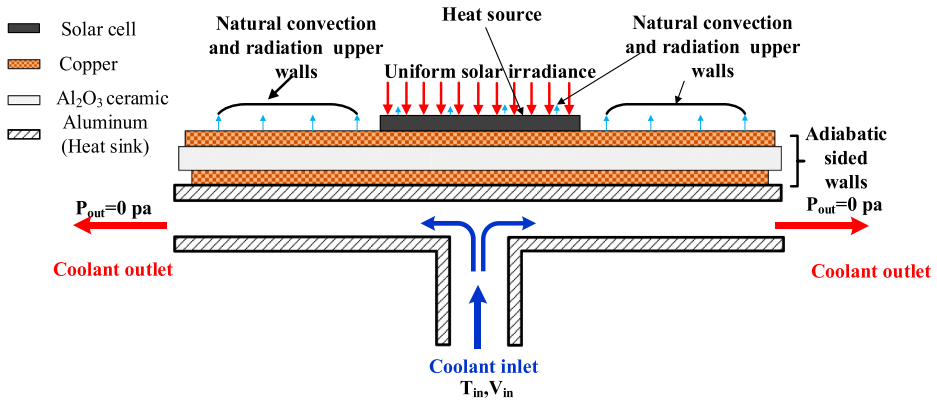


Fig. 4. Boundary conditions for 3J cell layers and HS assembly.

3.1.1. Triple-junction cell domain

The 3J solar cell layers conduction heat term can be as follows [38]:

$$\nabla\cdot(k_i\nabla T_i) + q_i = 0 \tag{1}$$

k_i, T_i and q_i denote the thermal conductivity, temperature and the solar irradiance absorption source term of the solid layer i. The equation for the Ge layer is as follows:

$$\nabla\cdot(k_{Ge}\nabla T_{Ge}) + q_{Ge} = 0 \tag{2}$$

The value of q_{Ge} is a function of concentrated solar light ($G = CR \times DNI$) [38]. In the current work, the $q_{Ge}$was determined as follows:

$$q_{Ge} = \frac{(1-\eta_{cell})\cdot G\cdot\alpha_{Ge}\cdot A}{V} \tag{3}$$

where η_{cell} and α_G is the cell's electrical efficiency and absorptivity respectively. The A and V represent the top surface area and volume of the cell respectively. The η_{cell}relation of the 3J cell under the operating T_{cell} can be formulated as follows [39]:

$$\eta_{cell} = \eta_{Ref} - [\beta_{Thermal}(T_{cell} - T_{Ref})] \tag{4}$$

where $\eta_{Ref} = 40.3\%$ when the reference temperature $T_{Ref} = 25\,^{\circ}C$ and $CR = 1000$ Suns [12,40], where $\beta_{Thermal}$ is the thermal coefficient and it is equal to 0.047% for 3J [17]. The value of the thermal coefficient is normally given by the cell manufacturer. However, the actual value of the temperature coefficient, in precise, is not subject to only on the material but on reference temperature as well. It is given by the following equation [41];

$$\beta_{Thermal} = \frac{1}{(T_c - T_{Ref})} \tag{5}$$

where T_c refers to the hight temperature at which point the cell electrical efficiency drops to zero. In the current study the value of $\beta_{Thermal}$ of the 3J cell of AZURESPACE product was equal to 0.047% [17].

3.1.2. Heat sink substrate

For the aluminum, the general equation of heat conduction was denoted by the following equation [42]:

$$\nabla\cdot(k_{sub}\nabla T_{sub}) = 0 \tag{6}$$

where the thermal conductivity of the aluminum substrate is k_{sub}.

3.1.3. Fluid domain and heat sink

The thermo-physical variation in water properties was considered using fourth order polynomial equations [43]. The governing equations for the fluid zone are as follows [17]:

Continuity equation:

$$\nabla\cdot(\rho_f\vec{V}) = 0 \tag{7}$$

Momentum equation:

$$\vec{V}\cdot\nabla(\rho_f\vec{V}) = -\nabla P + \nabla\cdot(\mu_f\nabla\vec{V}) \tag{8}$$

Fluid domain energy equation:

$$\vec{V}\cdot\nabla(\rho_f C_f T) = \nabla\cdot(k_f\nabla T) \tag{9}$$

where ρ is water density and μ, describes water viscosity. The subscript f refers to the fluid, $\vec{V}$ is velocity and P is the pressure of the coolant, respectively.

The previous equations were applied to calculate the 3D temperature, pressure, and velocity within all domains. Hence, the electrical powerP_{el}, friction power $P_{fric.}$, and net electrical powerP_{net} were calculated based on the equations exsit on [25]. The thermal power Q_{th} and the thermal efficiency (η_{th}) of the system are also defined [25].

3.2. Boundary conditions

The top surfaces were exposed to the heat losses by convection and radiation. At these surfaces, ambient temperature, convective heat transfer coefficient, exterior radiation temperature, and surfaces emissivity were defined. The inlet jet temperature and velocity were defined as uniform and normal to the inlet boundary. The pressure outlet was set as the zero-gage pressure boundary condition. Specifically, the boundary conditions for the HCPV/T are presented in Fig. 4 in more details.

3.2.1. The solid layer of HCPV

For the Ge layer top surfaces and the copper layer, we have the equations:

$$-k_{Ge}\left(\frac{\partial T_{Ge}}{\partial z}\right) = q_{rad,Ge\to S} + q_{conv,Ge\to a} \tag{10}$$

$$-k_{cu}\left(\frac{\partial T_{cu}}{\partial z}\right) = q_{rad,cu\to S} + q_{conv,cu\to a} \tag{11}$$

where the radiative $q_{rad,\ Ge\to s}$, and $q_{conv,\ Ge\to a}$ represent and convective heat transfer of the cell layer to the sky and ambient temperature, respectively. The equations of heat dissipation from the Ge were obtained using the following equations as an example [17]:

$$q_{conv,Ge\to a} = h_{conv,wind}(T_{Ge} - T_a) \tag{12}$$

$$h_{conv,wind} = 5.82 + 4.07V_{wind} \tag{13}$$

As $h_{conv,\ wind}$ represents the convective heat transfer coefficient by the wind. The radiative heat from the Ge was calculated as follows [17]:

$$q_{rad,Ge\to a} = \sigma\varepsilon_{Ge}(T_{Ge}^4 - T_s^4) \tag{14}$$

$$T_s = 0.0522T_a^{1.5} \tag{15}$$

The mixed boundary condition was defined with a precise definition of each parameter. For the interfaces between all adjacent solid materials, the accurate equations for the thermal coupling boundaries. For instance, Ge layer with the top copper interface, $T_{Ge} = T_{cu}$.

$$k_{Ge}\nabla T_{Ge} = k_{cu,I}\nabla T_{cu,I} \tag{16}$$

3.2.2. For heat sink design with coolant

The fluid inlet velocity $V_f = V_{in}$ and $T_f = 25\,^{\circ}C$. The fluid outlet pressure $P_{out} = 0$ with a backflow volume fraction of zero and backflow temperature of 25 °C were well defined.

For the heat sink, external walls were implicitly adiabatic walls. For the solid-fluid interface, no-slip in addition to thermal coupling were defined as follows:

$$v_f(x,y,z) = u_p i + v_p j + w_p k = 0,\ T_f = T_{Al}(x,y,z) \tag{17}$$

3.3. Exergy analyses

Exergy analyses were implemented in the current work to estimate the available energy, which is essential to exploit the wasted heat. The exergy analysis can help to identify where exergy losses arise and rank them according to their importance to provide an energy map for a more efficient systems. Normally, electrical energy is considered as pure exergy; the electrical exergy (E_{el}) is equal to the electrical power (P_{el}) [44]. The thermal exergy (E_{th}) was be obtained from the formula found in the literature [44,45]. In this mathematical model, the thermal exergy is equivalent to the thermal power times the Carnot factor:

$$E_{th} = \underbrace{\dot{m}\cdot c_f(T_{f,out} - T_{f,in})}_{thermal\ \ power}\cdot\underbrace{\left(1 - \frac{T_o}{T_{f,out}}\right)}_{Carnot\ \ factor} \tag{18}$$

Table 4
Structural properties of the cell and copper-I layers [48].

Property	Ge	Cu
Young's modulus, *E*, GPa	102.7	128
Shear modulus, *S*, GPa	40.1	47.05
Bulk modulus, *B*, GPa	77.78	152.4
Poisson ratio, ν	0.28	0.34
Coefficient of thermal expansions, α, 1/C°	6.1×10^{-6}	16.9×10^{-6}

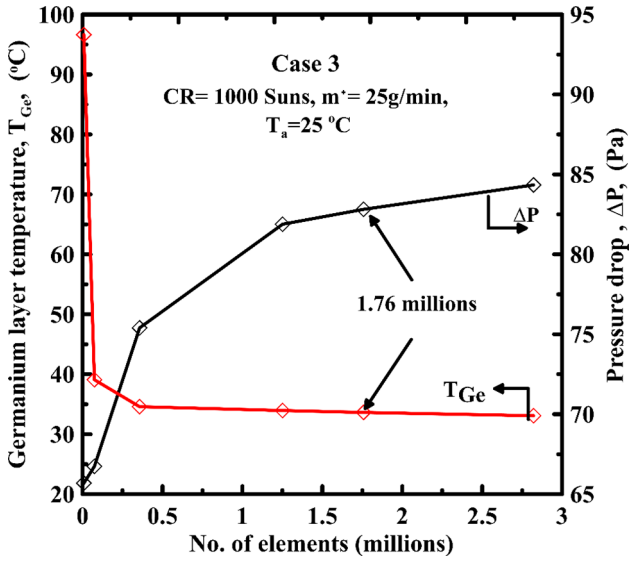


Fig. 5. Grid sensitivity test for the cell model and the HS in case 3.

where T_o is the reference temperature at which the no exergy content exists (dead state). and $T_{f,\ out}$ is the water outlet temperature.

For a liquid flow encountering heating processes from the inlet temperature $T_{f,\ in}$ to the $T_{f,out}$ as presented in this study, Evola and Marletta [44] reported that the Carnot factor term can be described as a function of the fluid mean temperature (T_m) using the following equation:

$$E_{th} = \underbrace{\dot{m}\cdot c_f\,(T_{f,out} - T_{f,in})}_{thermal\ \ power}\cdot\underbrace{\left(1 - \frac{T_o}{T_m}\right)}_{Carnot\ \ factor} \tag{19}$$

It is a standard view that solar radiation cannot be expressed as pure exergy [45]. Consequently, an appropriate conversion coefficient (Ψ_s) is essential and must be applied to the solar irradiance to calculate the exergy content of the solar radiation incident on the surface of the solar cell. The exergy conversion coefficient is described as follows:

$$\psi_s = 1 - \left(\frac{T_o}{T_s}\right) \tag{20}$$

where T_s is the sun solar radiation temperature. It is reported that T_s is equal to 6000 K [33]. Therefore, the value of the conversion coefficient considered in the present calculation is 0.95, as recommended by Petela [46] and Jeter [47] where $\Psi_s =$ (0.93–0.95). Then, the rate of exergy for solar radiation (e_{sol}) per unit surface W/m^2 can be calculated using Eq. (21):

$$e_{sol} = \psi_s \cdot DNI \tag{21}$$

Based on the previous model, the electrical, the thermal and the total exergy efficiency are ξ_{el}, ξ_{th}, and ξ respectively that are well defined in Evola and Marletta work [44]:

3.4. Structural model and thermal stress analysis

The application of MJ cells with complex multi-layer structures in terrestrial applications requires an in-depth understanding of thermo-mechanical reliability and assessment metrologies as the basis for enhanced lifetime predictions. The aging and lifetime assessment of the HCPV cells has been accelerated by stressing the solar cells in temperature. Hence, one of the main concerns of the current study is the assessment of the thermal stress in the interior of the 3J solar cell body. ANSYS mechanical was used to develop a finite-element analysis approach to understand the mechanical impact of temperature gradient and to estimate the thermal stresses. For simplification purposes and to reduce the computation time, the model considered the 3J solar cell and the copper-I substrate only. The calculated temperature distributions of the two layers based on the CFD model were imported as thermal loads to the structural model. The zero-stress condition was assumed at the reference temperature $T_{Ref} = 25$ °C. The back side of the copper-I was implicit as a fixed support, and all other faces were kept as load free surfaces. The mechanical parameters of each layer are displayed in Table 4.

As the 3J solar cell body was assumed to be a constrained body, it cannot move when a thermal load is applied. Subsequently, for a structural model, the structural simulation could be implemented by relying on the FEA concerning the principle of virtual work [38]. As the minimal change or the virtual in the internal strain energy (δu) should have the same variation in external work owing to the applied loads (δv). As the 3J solar cell body was motion constrained and in view of the strain energy attributed to the thermal stresses resulting from the thermal load, the execution of virtual work principle under a temperature body load yields are detailed in Appendix A.

3.5. Solution methodology

The comprehensive solution procedure was implemented for the coupled thermal and mechanical models. After generation of the geometry with the dimensions presented in Tables 1–3, a mesh independence study was performed. The T_{cell} and pressure drop across the HS were obtained to perform the grid sensitivity test. After selection of a suitable mesh test, the CFD model was iteratively simulated. Specifically, in the first iteration ($S = 0$), the T_{cell} and efficiency were set to the reference values. At that time, the generated heat value was calculated using Eq. (3). For each new iteration, new T_{cell} and efficiency values were obtained, and a new heat generation value was considered using Eq. (5). This step was repeated till the difference between the benefits of the last two calculations of the T_{cell} was less than 3%. Then, the CFD model was coupled with the structure model. At that time, the structural model was well defined to compute the mechanical stresses based on the thermal load which arises due to the temperature distributions inside the HCPV/T.

In the case of the HS model, the mesh independance test was studied by estimating the pressure drop across the HS fluid domain. For illustration, Fig. 5 shows the test results predicted for the Case 3 design. It is obvious from Fig. 5 that the grid was approximately independent at the fifth grid size. For Case 3, a total number of 1,757,976 elements were selected to compromise between the calculation accuracy and the computational time. The computational domain was divided into multi-zones and thermally coupled together. This idea allows us to control the size of each zone separately to increase the mesh density in the fluid zones while decrease it in the solid zones to compromise between the solution accuracy and the computational time. The mesh details are displayed in Fig. 6 for case 3 as an example.

4. Validation

The model was authenticated by comparing it with the existing mathematical and empirical work in the literature, as represented in

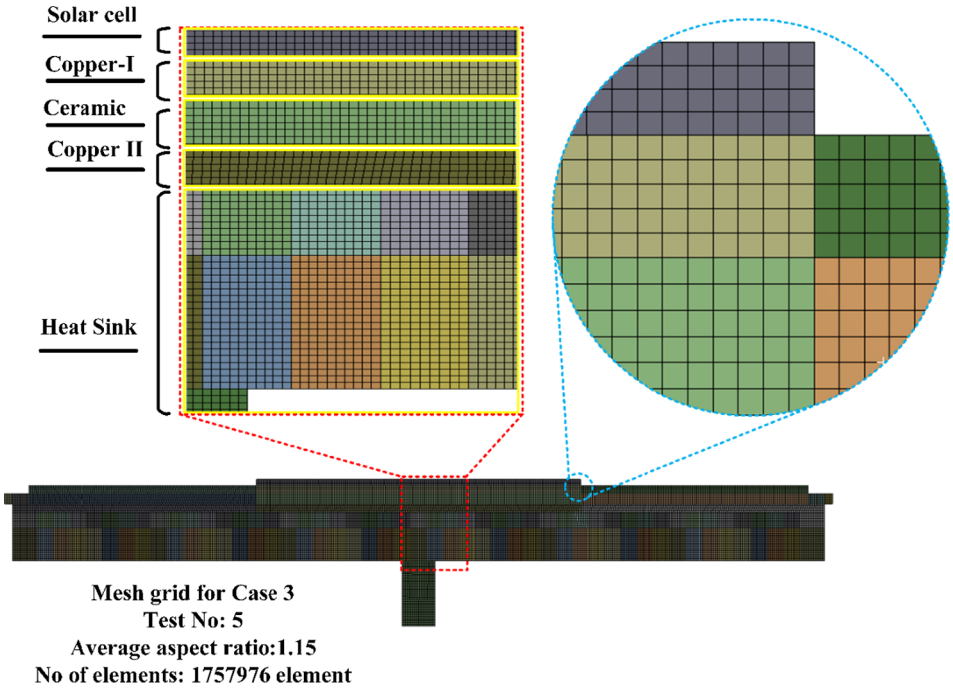


Fig. 6. Grid structure for cell model and HS in case 3.

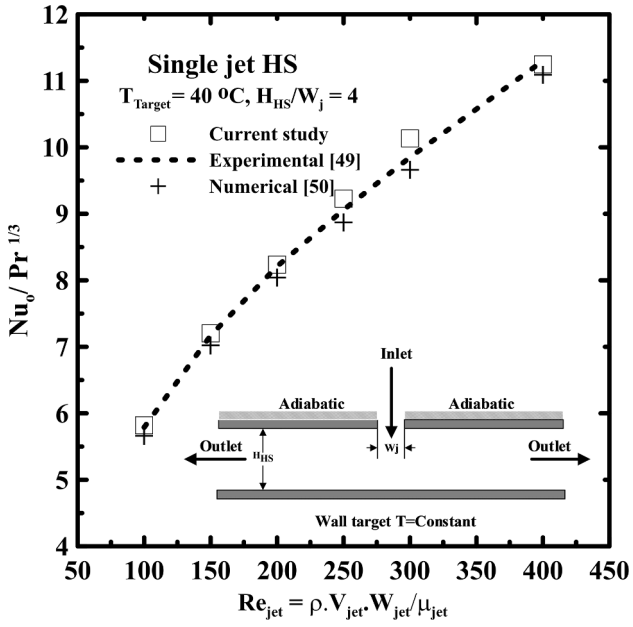


Fig. 7. Comparison of current study data the literature empirical [49] and numerical [50] results.

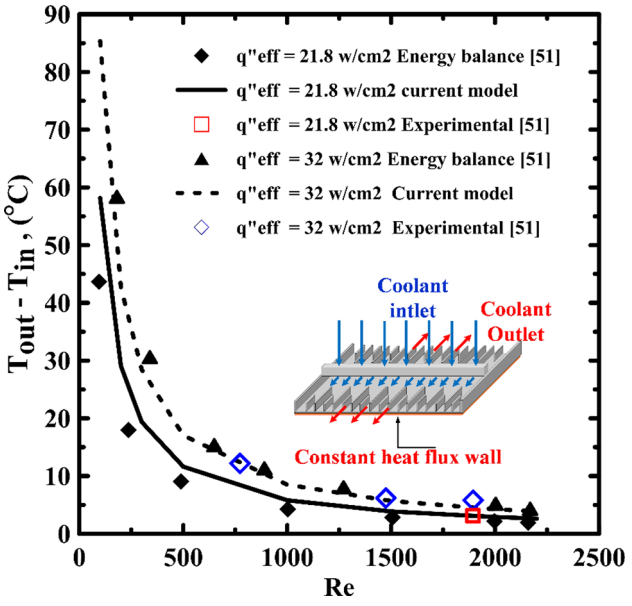


Fig. 8. Comparison of results of the current model with the literature for empirical and numerical work [51] regarding $\Delta T = T_{out} - T_{in}$.

Fig. 7. The heat transfer characteristics JI heat sink was validated with the empirical data of Hung et al. [49], at the same time the predicted results were compared with Manca et al. [50] mathematical data. In particular, the stagnation point Nusselt number (Nu_o) was in a satisfactory arrangement with the empirical correlation and the numerical results. The assessments showed an insignificant variation concerning the Nu_o when $H_j/W_{HS} = 4$ when the Reynolds number (Re) varied from 50 to 400. Certainly, the average error between the current model and empirical data was about 1%, and the maximum variation was equal to 2.9%. At the same time, the average error between current model results and mathematical results was about 3%, and the maximum deviation was equal to 4.9%.

As shown in Fig. 8, the single inlet hybrid JIMC cooling scheme was validated by comparing them with the existing CFD and experimental data in the literature [51]. The results based on the current model were in acceptable agreement with the empirical and numerical results of Barrau et al. [51]. The comparison between the present predicted results and those obtained by Barrau et al. was carried out in term of fluid

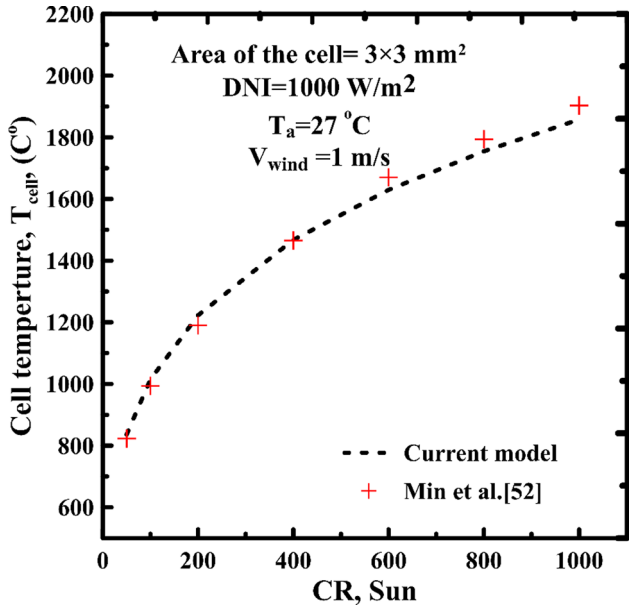


Fig. 9. Validation of the current predicted results with data of Min et al. [52].

Table 5
Validation of the current predicted results, T_{cell} (Pred.), with the data in [53].

	T_a = 25		T_a = 35		T_a = 45	
h_{Back}	T_{cell} (Pred.)	T_{cell} [53]	T_{cell} (Pred.)	T_{cell} [53]	T_{cell} (Pred.)	T_{cell} [53]
1200	75	76	85	87	95	98
1300	72	73	82	83	92	94.5
1400	69	70	79.	80.5	89	91.5
1500	67	68	77	78	87	89
1600	65	66	75	76	85	86.5

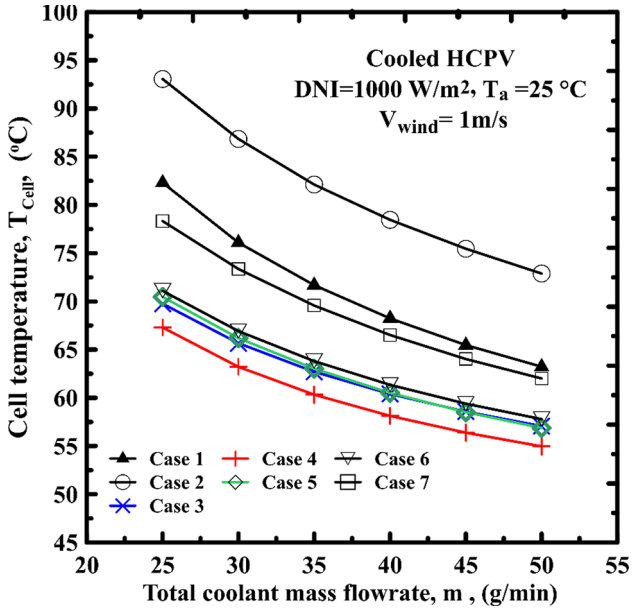


Fig. 10. Variation in T_{cell} with water $\dot{m}$ for all heat sink cases.

outlet and inlet temperature difference ($\Delta T = T_{out} - T_{in}$) with *Re* of 50–2200. The predicted results were in satisfactory agreement with the mathematical and experimental data at all given Re where the average relative variation between numerical and experimental results of the ΔT was less than 14% and 5%, respectively

The thermal model of the HCPV system was validated with another two sets of works existing in the literature. The first, the 3J solar cell thermal model was validated and compared with the numerical data of Min et al. [52] as shown in Fig. 9(a). The comparison between the temperature values of the 3J cell was estimated for a $3 \times 3\,m^2$ cell for different CR values. The current model calculations agreed with Min's results with a minor error of 1.9%.

In the second set of validation, the predicted results were compared with the results in [53]. In this simulation, the solar cell loses heat from its upper surface to the surrounding ambient via combined convection and surface radiation. The back-side (copper II substrate) dissipate heat to the surrounding by surface radiation and convection, where the backside convective heat transfer coefficient ranged from 1200 to 1600 $W/(m^2\,K)$ and the ambient temperature, T_a, was kept constant value at three different values of 25, 35 and 45 °C. The comparison between the predicted results and the results in [53] showed a satisfactory matching between the two models with an error of 2% (as seen in Table 5).

5. Results and discussion

The average T_{cell} with a change in the inlet coolant $\dot{m}$ for all the studied cases is shown in Fig. 10. It is noticeable from Fig. 10 that the T_{cell} declined as the $\dot{m}$ of the coolant was augmented. For $\dot{m}$ = 50 g/min, a significant reduction in the T_{cell} for all cases was detected. The JIMC scheme of the investigated cases (Cases 3–7) achieved a lower T_{cell} than the JI scheme configurations along the $\dot{m}$ range. For the conventional JI cooling scheme, the thermal resistance remained constant lengthwise along the flow direction. Furthermore, water temperature raised along the flow stream because of the heat transfer from the inlet to the outlets of the HS. Consequently, the heat transfer considerably dropped when the impingement zone was destroyed. For the hybrid design cases, when the numbers of microchannels increased, their hydraulic diameter decreased. Consequently, the drop in the flow stream section allowed the flow velocity to be increased. Meanwhile, the heat transfer zone was stretched as confirmed by Tuckerman and Pease [54]. Moreover, the absent of the cross-flow increased the heat dissipation rates from the cell body.

As previously discussed, many factors attributed to the degradation of the cell performance such as current mismatching [55]. Current mismatching due to a non-uniformly distributed T_{cell} and hot spots on the cell leads to short-term losses that are highlighted by a reduction of the efficiency of the cell or long-term effects such as failure due to thermal stresses [56]. To determine the local T_{cell} distribution, the T_{cell} contours are presented in this section. The worst-case condition (no cooling situation) under different *CR* values was investigated in an earlier work of the authors [31]. For an uncooled solar cell under *CR* = 1000 Suns, the authors reported that the local T_{cell} a reached up to 1360 °C with a T_{cell} non-uniformity of up to 16.9 °C. For the cooled cell, the contours of the local T_{cell} with JI scheme cases for the lowest and highest $\dot{m}$ = 25 and 50 g/ min respectively, are presented in Fig. 11. For $\dot{m}$ = 25 g/min, the highest local temperature was declined from 1360 °C to about 83.7 °C and 98.4 °C for Case 1 and Case 2,

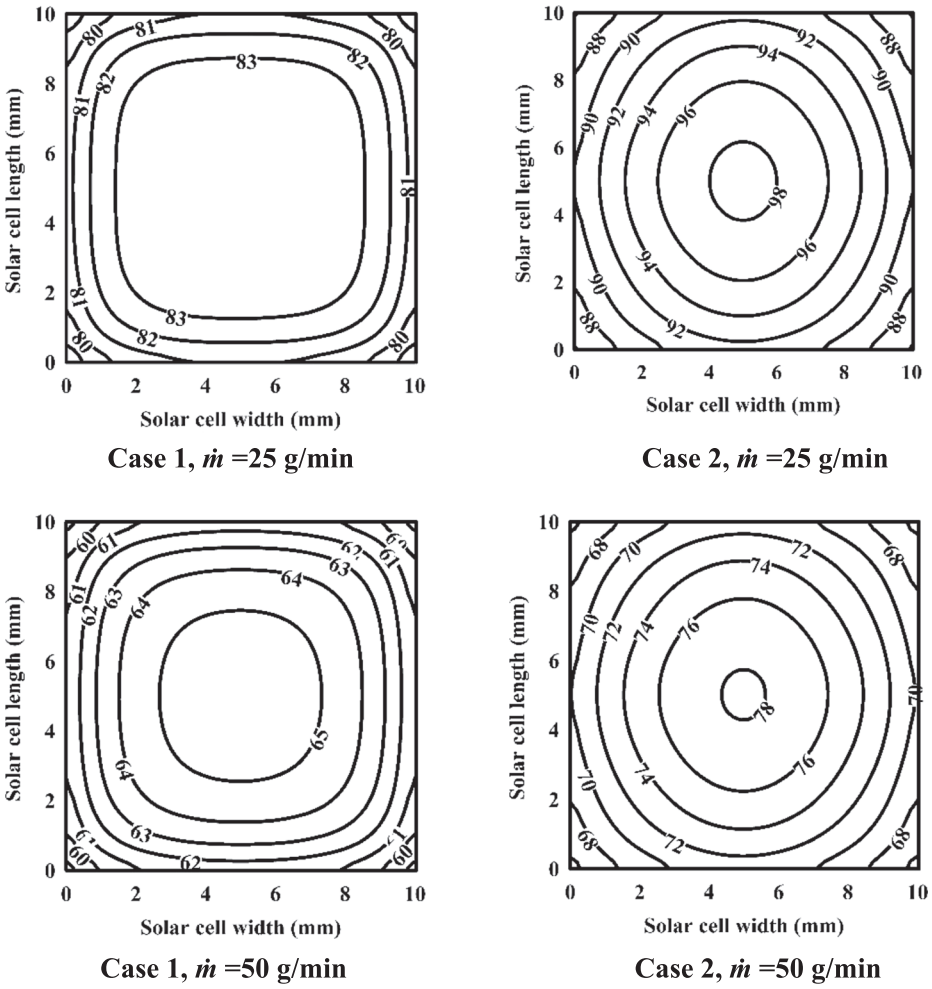


Fig. 11. Local T_{cell} contours over the surface of the cell layer for $\dot{m}$ of 25 and 50 g/min within JI scheme configuration.

respectively. Based on these contours, when the inlet flow rate was increased, the local temperature of the cell significantly declined. For the highest $\dot{m}$ the minimum T_{cell} was 65.6 °C and 78.2 °C for Case 1 and Case 2, respectively.

The contours of local T_{cell} with hybrid cooling HS scheme cases are shown in Fig. 12. It can be observed from these contours that the temperature distribution is extremely affected by the channel design and flow distribution with the HS. For the normal JI schemes, the maximum local temperature occurs at the cell center. Furthermore, the lowest T_{cell} is observed at the cell corners area, as indicated by the contours. For instance, it is evident from the hybrid scheme contours that for $\dot{m}$ = 25 g/min, the highest local T_{cell} decreased considerably to 71 °C, 69 °C, and 79 °C for Case 3, Case 4, and Case 7, respectively. Based on these contours, when the for $\dot{m}$ was augmented from 25 g/min to 50 g/min, the cell's local temperature decreased noticeably in the range between 12 and 16 °C. When the $\dot{m}$ was 50 g/min, the highest local T_{cell} values were about 59, 57, and 63 °C for Case 3, Case 4, and Case 7, respectively. The comparison between the conventional JI and hybrid schemes demonstrated the considerable reduction of the average T_{cell} and local temperature distribution when the hybrid cooling scheme was applied.

In particular, the T_{cell} non-uniformity which is the difference between the maximum and minimum cell temperatures ($\Delta T = T_{Max} - T_{Min}$). The ΔT was calculated and compared as shown in Fig. 13. It is clear that the $\dot{m}$ had a slight effect on the ΔT when a comparison is made between Fig. 13(a) and (b). For instance, Case 7 has the lowest ΔT values of 5.9 °C and 5.3 °C for the $\dot{m}$ = 25 g/min and 50 g/min, respectively. For Case 1, the ΔT was approximately 6 °C for a $\dot{m}$ = 25 g/min and 7.5 °C $\dot{m}$ = 50 g/min. The most significant ΔT was 16.9 °C for the uncooled cell.

Fig. 14 shows the different temperature and velocity contours of Case 1 when $\dot{m}$ = 50 g/min. Fig. 14(a) reveals the velocity contour at a plane located at the middle of the heat sink width (B-B at Fig. 14(c)). It is obvious that the impingement zone developed at the center of the impinging target surface of Case 1 and the thin boundary layer was extended at the surrounding area of the impinging zone. The highest velocity the center of HS where a then boundary layer occurred at that zone, consequently high heat transfer augmented directly underneath

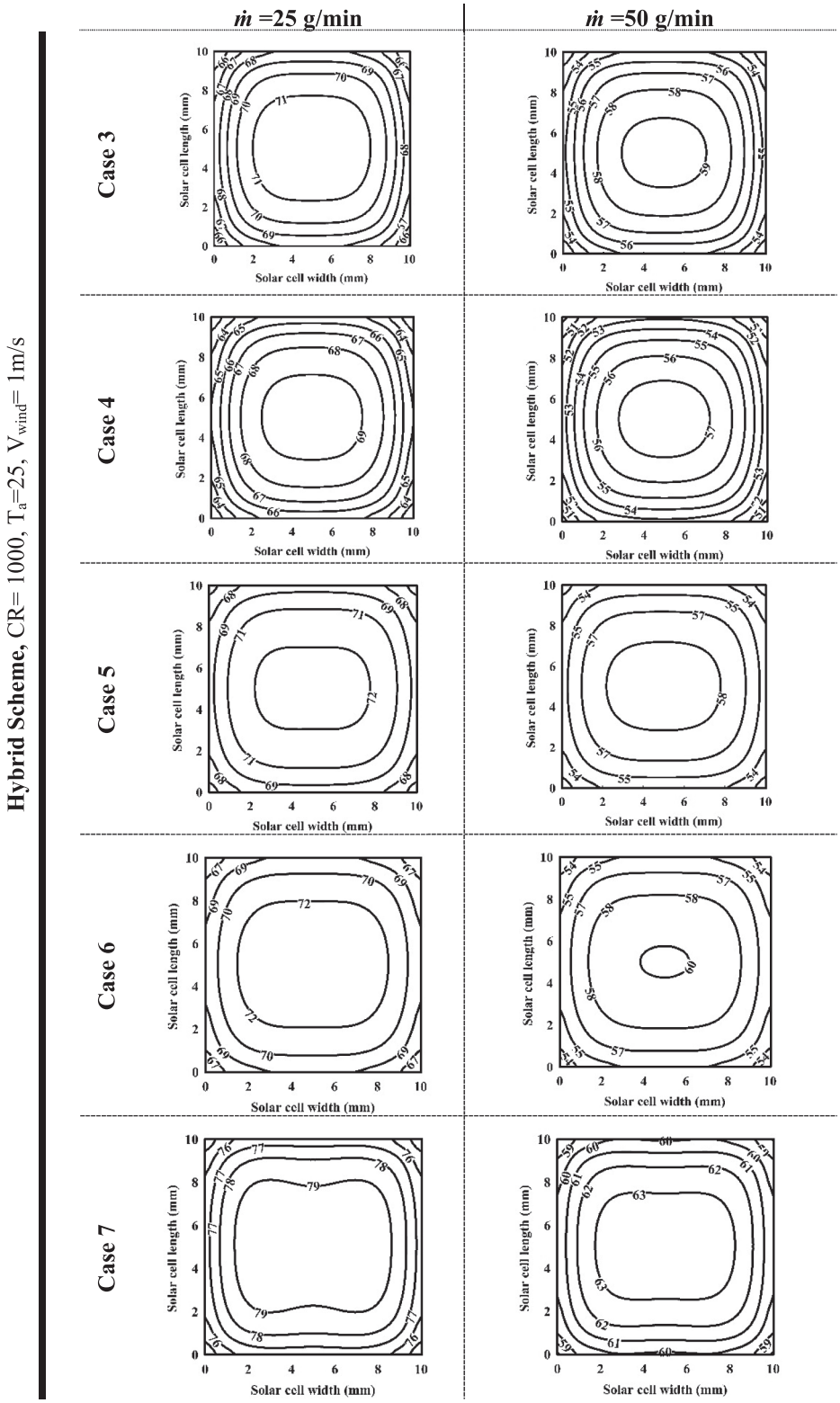


Fig. 12. Local T_{cell} contours over the cell surface for $\dot{m}$ = 25 and 50 g/min within hybrid scheme cases.

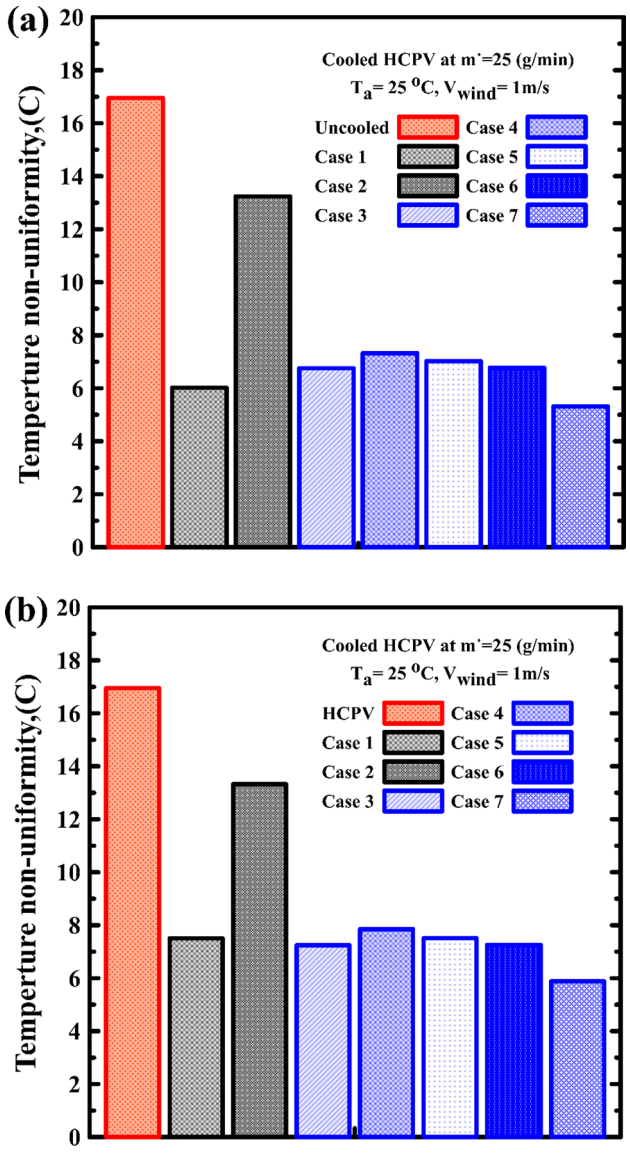


Fig. 13. Non-uniformity of T_{cell} for uncooled cell and the all investigated cases for (a) $\dot{m}$ = 25 g/min and (b) $\dot{m}$ = 50 g/min.

the cell location. The velocity gradually decreased afterward the impinging zone consequently the heat transfer. Fig. 14(b) illustrates the temperature contour for the same plane in Fig. 14(a). The temperature showed a similar contour pattern as the velocity. From the contours, it is obvious that the highest temperature occurred at the cell body specially at the cell central part. For more details, velocity contour at a plane in the middle of the heat sink height (A-A location, Fig. 14(a)) shows the velocity distribution over the heat sink different counter-rotating vortex structures were formed as the jet impinges restraining effects on the bottom surface and only one jet impinging point. Fig. 14(d) shows the temperature contour for the target surface (Aluminum layer back side) at $\dot{m}$ = 50 g/min. Even the high heat dissipation at center the local temperature was high in that area this because of the highest heat flux derived from the solar cell. Relying to that figure, the jet impingements cooling scheme could not be a stand-alone cooling scheme for HCPV under high *CR* values. Further progress must be toward advance the JI cooling schemes in that case.

The velocity contours for a plane in the middle height of the heat sink (A-A location) for the six remaining cases at the highest $\dot{m}$ are shown in Fig. 15. For Case 2 four inlet jets were used instead of a single jet, the inlet velocity from each jet was low when it is compared with Case 1 as a result of the same inlet mass flow rate was applied for all cases. Consequently, the jet impinging effect was insignificant in case 2 and the cooling capacity of this design was restricted as will be discussed later. The effect of the impingement zone with involving microchannels in Case 3 to Case 7 is clear in Fig. 15. The effect impinging free stream zone in addition to the channels orientation in these cases let to a variation in velocity distribution in the fluid domain. The influence of the variation had a significant effect in local cell temperature subsequently the cell temperature uniformity as illustrated in the temperature contours in Fig. 12 and bar charts in Fig. 13.

As the average T_{cell} declined, the cell electrical efficiency was enhanced as proved in Fig. 16. It is evident that Case 4 of the hybrid cooling scheme resulted in higher electrical efficiency of the cell along the inlet cooling $\dot{m}$ range. Case 2 recorded the lowest cell electrical efficiency values. For $\dot{m}$ = 50 g/min, the maximum cell electrical efficiency of 39.7% was recorded for Case 4. The effect of the $\dot{m}$ of the coolant on the thermal power of the HCPV/T system is revealed in Fig. 17. It is obvious that the $\dot{m}$ increasing resulted in a decline in the thermal efficiency of the HCVP/T arrangement. A similar trend was exhibited for all the investigated designs. This mainly occurred as the electrical efficiency increase and the percentage of the solar energy transformed into useful thermal energy declined, the part of the collected heat decreased. For illustration, Case 7 achieved the highest thermal efficiency of 60.4%. Simultaneously, the electrical efficiency had a value of 39.3% for the lowest $\dot{m}$ value.

A comparison between Figs. 18 and 19 shows the contrariwise proportional relationship between the electrical and thermal efficiency at various investigated $\dot{m}$ values for all cases. Fig. 18 shows the behavior of the cell electrical power with respect to $\dot{m}$. All the investigated designs exhibited an increase in P_{el} as $\dot{m}$ increased. Note that at $\dot{m}$ = 50 g/min, the electrical power output achieved its maximum value for all the investigated HS designs. Case 4 had the greatest electrical power output value for the entire $\dot{m}$ range with electrical power output values of P_{el} = 33.6 and 33.7 W at $\dot{m}$ = 25 and 50 g/min, respectively. Case 2 exhibited the lowest P_{el} values for all $\dot{m}$ range, for instance P_{el} = 33.2 and 33.5 W at $\dot{m}$ = 25 and 50 g/min, respectively. However, the gain in thermal energy (P_{th}) exhibited a slightly decrease for the hybrid scheme as the $\dot{m}$ increased. In the case of JI, the reduction was more pronounced. For example, as shown in Fig. 19, Case 2 P_{th} = 51.2 and 51.1 W at 25 g/min and 50 g/min, respectively. For Case 4, P_{th} = 51.2 and 50.8 W as the $\dot{m}$ increased from 25 to 50 g/min because of the increment in the electrical power. It is evident from the Fig. 19 that the highest and lowest gained in the thermal power were attained for Case 7 of the hybrid scheme and Case 1, respectively.

The exergy analyses provide a well-organized arrangement design and promotes electricity generation. As such, this result is valuable for research in the field. The typical exergy efficiency for electricity is higher than the thermal energy recapture [44]. Fig. 20(a) and (b) shows the change in the electrical and thermal exergies with the $\dot{m}$ values, respectively. At this point, it should be noted that the E_{el} and E_{th} showed

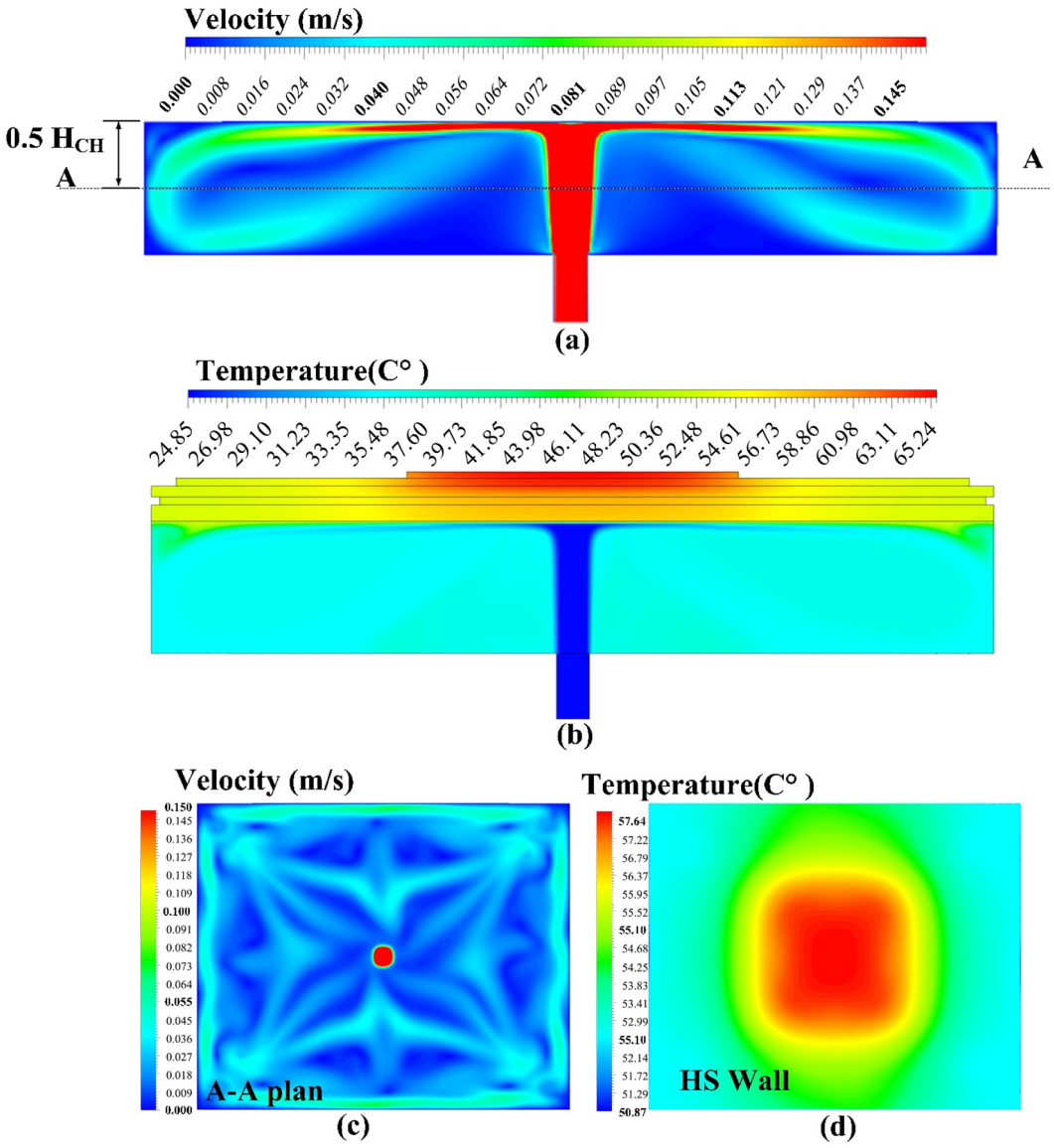


Fig.14. Case 1 contours when $\dot{m}$ = 50 g/min of (a) velocity and (b) temperature HCPV layers at a plane in the middle of the heat sink width. (c) velocity for a plane in the middle of the heat sink height and (d) temperature at the impinging surface at the same $\dot{m}$.

an analogous trend of η_{el} and η_{th}. In addition, the output electricity E_{el} was lower than its η_{el}, although the P_{el} was confirmed as pure exergy. The main reason is the ψ_s the coefficient that was considered and calculated using Eq. (29), which was approximately 0.95.

As the $\dot{m}$ of water was increased, the ξ_{el} increased. In contrast, the ξ_{th} was reduced with an increase in $\dot{m}$. The reduction in the ξ_{th} at high flow rates can be explained by two main reasons. Firstly, a small temperature variation among inlet and outlets of the coolant high flow values caused a very low exergy content. Secondly, as discussed previously, the high electrical efficiency leads to a low-grade heat energy. For rejected heat exploitation, it is suggested that the HCPV/T be operated with low flow rates. Moreover, it is clearly shown in Fig. 21(b) and (c) that the investigated cooling schemes exhibit a minor difference in the ξ_{th} and ξ. Consequently, it could be assumed to be the same at each inlet cooling flow rate.

This variation in thermal exergy is a consequence of the change in coolant outlet temperature. From Eqs. (12) and (28), the temperature difference between the outlet and inlet temperature of the coolant $(T_{f,out} - T_{f,in})$ term is relatively close for all investigated cooling schemes at the same inlet mass flow rate, the Carnot factor contains $T_o/T_{f,out}$ is the temperature ratio between system's dead state temperature (no exergy contents at T_o = 25 °C) and the $T_{f,out}$, hence the Carnot factor was quite similar also. Formerly, $T_{f,in}$ was assumed to be constant$T_{f,in}$ = 25 °C and the simulation work was implemented for all cases under the same condition of $\dot{m}$, CR and DNI. Hence, the key parameter was the change in $T_{f,out}$ which was insignificant. In the other

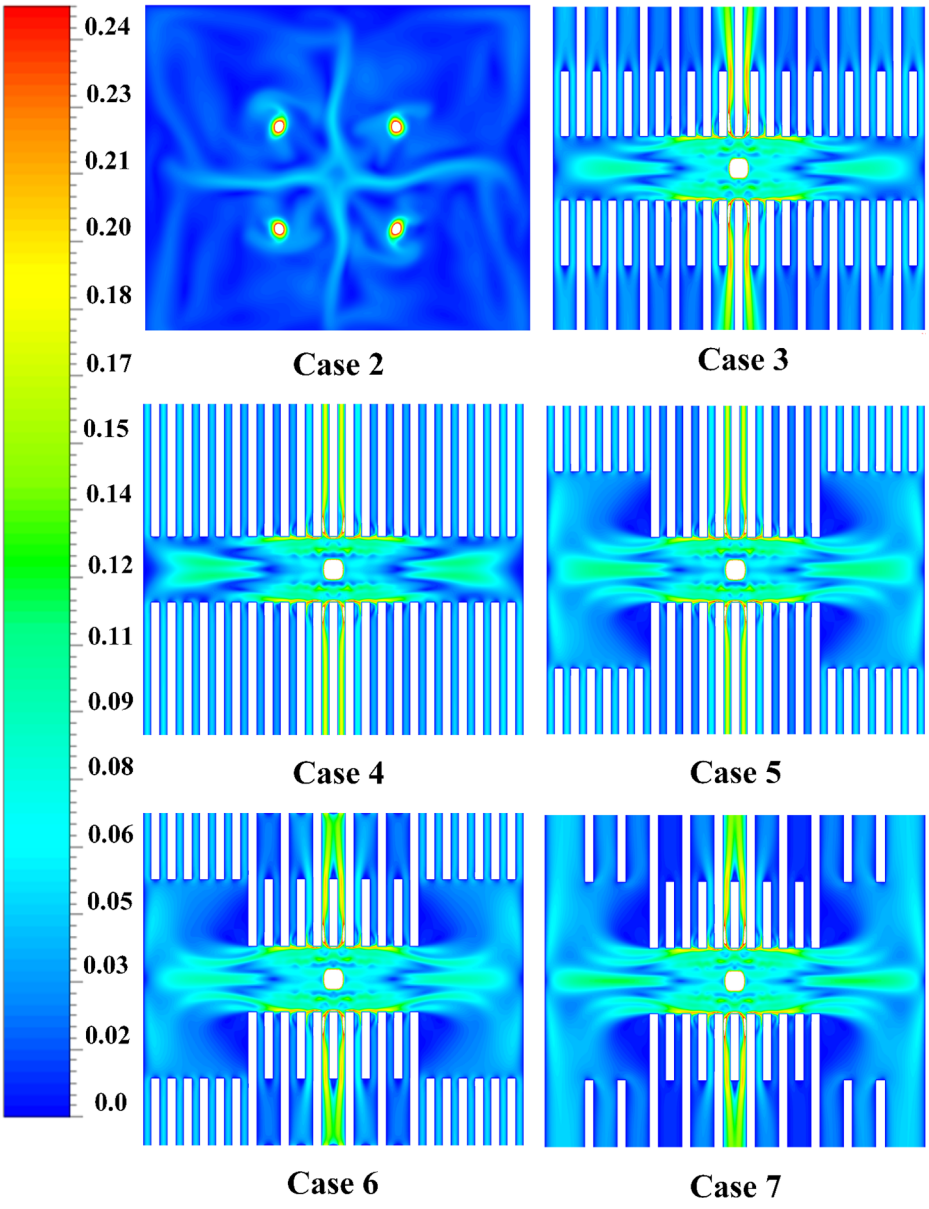


Fig.15. Velocity contour for a plane in the middle of the heat sink height $\dot{m}$ = 50 g/min for Case from 2 to 7.

hand, the variation on the inlet mass flow rate had a great effect on the thermal exergy of all the studied designs. Henceforth, it is hard to notice the variation on each case from the figure.

Therefore, to exploit the removed heat, it is suggested that the HCPV/T be operated with low flow rate values. Nevertheless, the exergy evaluation must be extended for an extensive range of the cooling flow rates under two-phase flow condition. For the overall exergy efficiency ξ, the sum of the ξ_{th} and ξ_{el} exergies at the maximum value is equal to 53.5% at the lower flow rate of 25 g/min for Case 3, as shown in Fig. 20(c). From the above discussion, it is concluded that the thermal and total exergy loss of the HCPV/T system can decline to some extent by the reduction of cooling water $T_{f,out}$ from HCPV/T [33]

One of the important parameters of the thermal analysis is pressure drop, which may negatively affect the overall efficiency of the system by reducing the net power gain. As the pressure drop rises, the total efficiency will decline. For this investigation, when the $\dot{m}$ was increased, the pressure drop augmented for all the studied designs. Consequently, there was an increase in the friction power. It was determined that Case 1 had a higher pressure drop across it when compared with the all other designs, as shown in Fig. 21. It is worth noting that all the studied designs showed the benefits of low friction losses, including Case 1. These losses may be ignored. For the hybrid schemes, a pressure loss was exhibited between Case 1 and Case 2 for all the studied cases. All the hybrid scheme cases had almost the same values

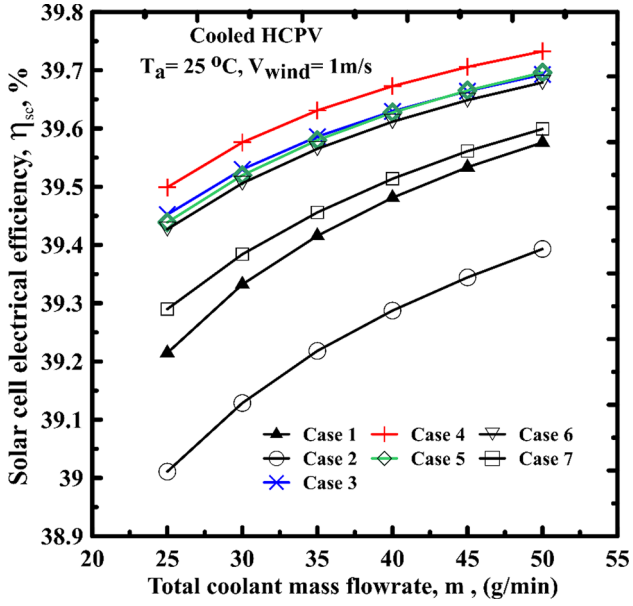


Fig. 16. Variation in cell electrical efficiency with $\dot{m}$ for all cases.

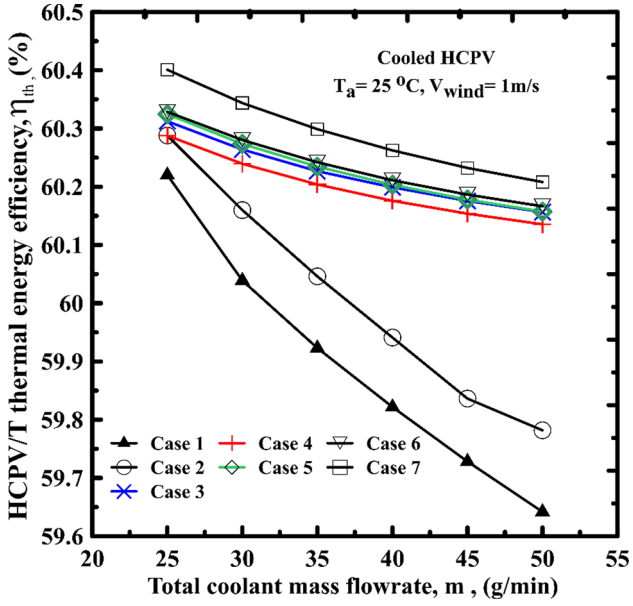


Fig. 17. Variation in the thermal efficiency with $\dot{m}$ for all Cases.

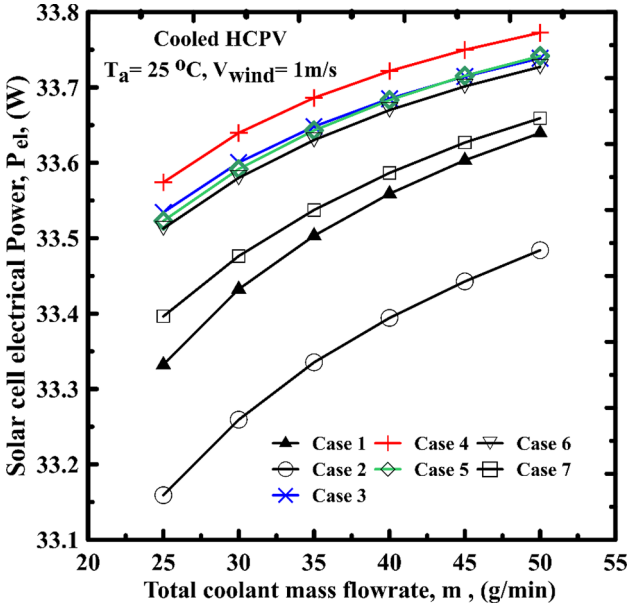


Fig. 18. Variation in electrical power of the cell with $\dot{m}$ for all cases.

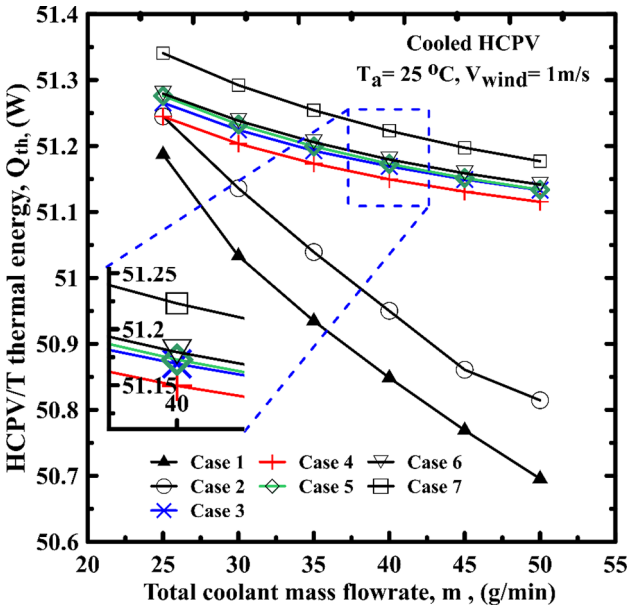


Fig. 19. Variation in HCPV/T thermal power with $\dot{m}$ for all cases.

as shown in the figure.

The economic feasibility assessment and life prediction of the of HCPV modules could not be performed without an in-depth understanding of the mechanism of thermal cycling. The structural model was used to estimate the thermal stresses based on the material mechanics theory of the HCPV cell. It was noticed from the thermal stress anlysis on the cell top surface for both for all studied designs that, the thermal stress distribution is analogous to temperature distribution within the cell body by means of the highest local thermal stress was located at the center for all cases. It was detected from the thermal stress analysis, that at $\dot{m}$ = 25 g/min, the highest local thermal stress was equal to 51.5 MPa and 64 MPa for Cases 1 and 2, respectively. When the inlet flow rate was increased, the local thermal stress of the cell reduced considerably. At the $\dot{m}$ = 50 g/min, the maximum thermal stress was approximately 35.3 MPa and 45.7 MPa for Case 1 and 2, respectively. For, Case 4 had the lowest local temperature and the lowest thermal stresses. The thermal stress reduced with any rise in the $\dot{m}$ value. For instance, the higher thermal stress at the top surface of Case 4 decreased from 38.8 to 28.8 Mpa when $\dot{m}$ was increased from 25 to 50 g/min. The comparison between the thermal stress of the hybrid and JI cooling schemes of the solar cell showed a considerable reduction for the average thermal stress of the cell with the hybrid cooling

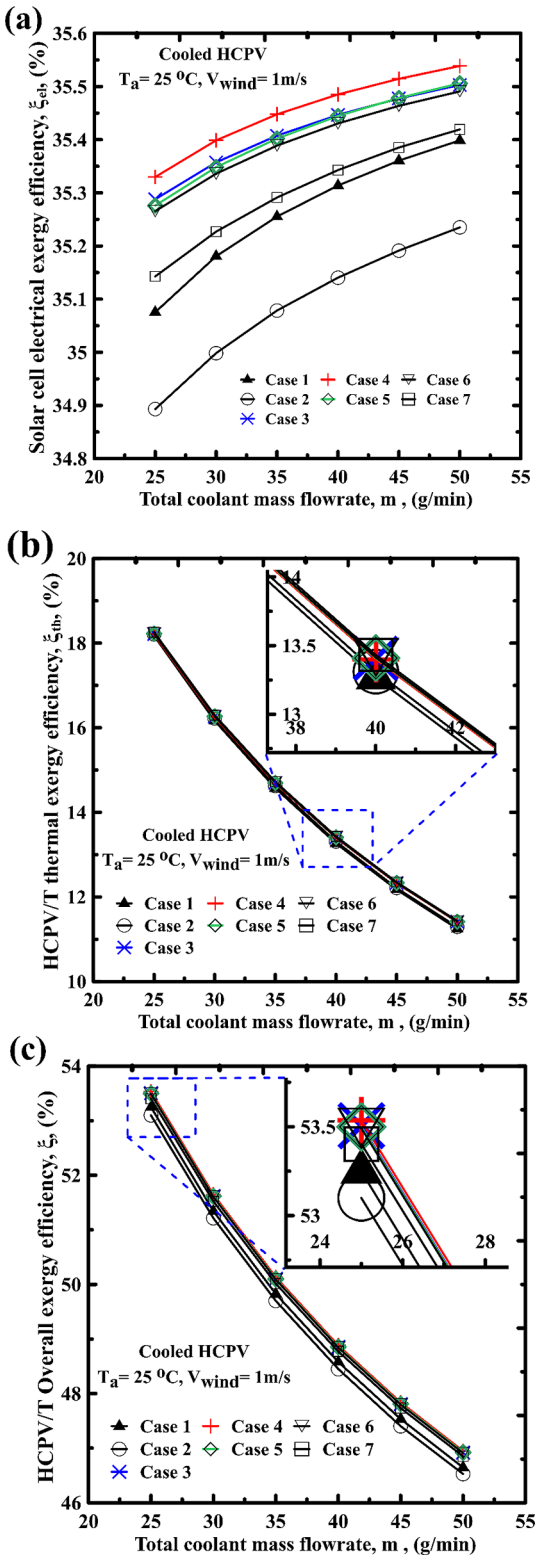


Fig. 20. Variation in (a) electrical, (b) thermal, and (c) overall exergy efficiencies with the change in $\dot{m}$ values.

scheme. It is evident that the maximum and the lowest thermal stress are in the center and the edges of the cell body respectively, for all the investigated cases. This investigation is the first step in gaining an in depth understanding of the structural integrity of HCPV modules and is thus beneficial to the solar energy community. Finally, these results are very useful for extended characterization techniques. They may make

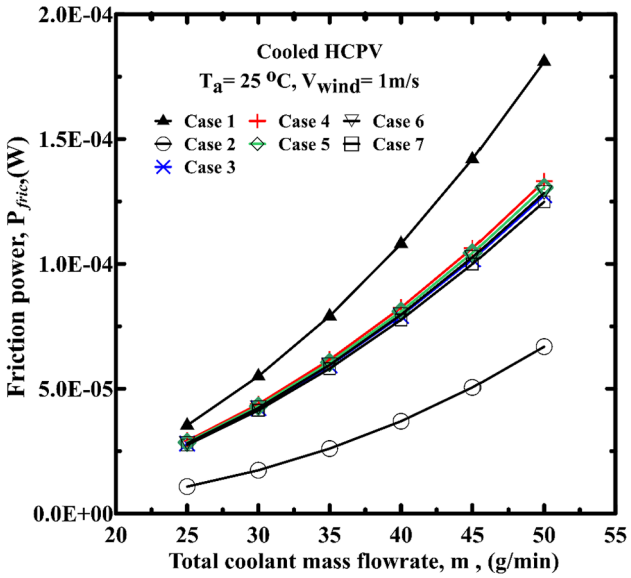


Fig. 21. Effect of $\dot{m}$ values in the pressure drop between the inlet and outlet of all the heat sinks.

available insight not only into the behavior leading to thermal stress on the top cell, which normally leads to cracking failures in MJ cells, but also denote a blueprint for improving gridline materials to reduce the capital costs of these systems, and the upgrading of more efficient HCPV arrangements.

6. Conclusions

The present work provides an important step to reach the optimum cooling design conceptions for HCPV systems and to improve its electrical yield under high concentration ratio. The comprehensive comparison between jet impingement/microchannel and jet impingement was performed for the purpose to keep a multijunction solar cell under the desired operating conditions under high solar irradiance concentration of 1000 suns. For the hybrid designs, the effect of channel width variation and impingement zone was figured out. The results of this study show that the hybrid cooling scheme is effective and it can attain the recommended operation temperature (50–80 °C) under high solar concentration ratio of 1000 suns where the solar cell temperature remains 55 °C with minimal pumping requirements. The hybrid cooling schemes attained 0.86% higher electrical efficiency of HCPV than conventional jet impingements. The jet impingement/microchannel hybrid scheme attained a satisfactory uniformity in cell temperature with the lowest temperature nonuniformity of 5.8 °C. Moreover, the results showed that the highest exergy contents were achieved when the HCPV was cooled with low flow rates. The studied cooling schemes for HCPV can operate with a negligible friction loss. In conclusion, the implemented approach could be useful in enhancing the overall performance of the HCPV module. This study has elucidated that further challenge for more efficient use of the HCPV system is a development of a system for exploiting the rejected heat. To realize the proposed cooling schemes, further investigations must be conducted based on the economic point of view.

Acknowledgements

Sincere gratitude and appreciation for the Egyptian Ministry of Higher Education (MOHE), EJUST University and JICA for providing the opportunities and offering the support and for this manuscript. This work was supported by the Science and Technology Development Fund, Egypt [Project ID: 33515].

Appendix A. Auxiliary equations for the mechanical model

The Virtual work principle under a temperature body load yield

$$\{\delta u\}^T \int_{vol} [B]^T [D][B] dv \{u\} = \{\delta u\}^T \int_{vol} [B]^T [D]\{\varepsilon^{th}\} dv \quad (1)$$

where $\{\delta u\}^T$ is the combination of the virtual arbitrary displacement vector for all the above terms, $\{\varepsilon^{th}\}$ is the thermal strain vector, [B] is the displacement gradient matrix and [D] is the material properties matrix [13,38]. Then, previous equation can be rewritten as follows;

$$[K]\{u\} = \{F^{th}\} \quad (2)$$

where [K] represents the element stiffness matrix,$\{F^{th}\}$ is the element load vector and $\{\varepsilon^{th}\}$are described in the following equations [57];

$$[K] = \int_{vol} [B]^T [D][B] dv \quad (3)$$

$$\{F^{th}\} = \int_{vol} [B]^T [D]\{\varepsilon^{th}\} dv \quad (4)$$

$$\{\varepsilon^{th}\} = \{\alpha\}\Delta T \quad (5)$$

The thermal stresses produced in the cell body can be calculated considering Hook's law using the following equation [57];

$$\{\sigma\} = [D]\{\varepsilon^{el}\} \quad (6)$$

where $\{\sigma\}$ and $\{\varepsilon^{el}\}$represent the stress vector and the elastic strain vector, respectively. The strain$\{\varepsilon\}$ is equal to the sum of the elastic and thermal strains as follows;

$$\{\varepsilon\} = \{\varepsilon^{el}\} + \{\varepsilon^{th}\} \quad (7)$$

The cell body was constrained; therefore, the total strain is equal to zero. Therefore, the thermal stresses could be estimated by rewriting Eq. (6) as follows;

$$\{\sigma\} = [D]\{\varepsilon^{th}\} \quad (8)$$

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